

Burger AI Kitchen

A Layered Metaphor for Training, Alignment, Recalls, and Invariants

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1 Overview

This document sketches the “Burger AI” metaphor: an end-to-end picture of how to design, train, align, monitor, and recall AI agents using the language of a restaurant kitchen.

The framing goal is simple:

People want something specific from AI systems. Instead of pretending one generic tool can please everyone, design a kitchen where many burgers (agents) can be made consistently, adjusted safely, and retired responsibly.

Key ingredients:

- A layered burger stack that maps cleanly onto model internals.
- Kitchen hardware: grill, press, marinade tank, sauce bar, fire suppression, pass.
- A recall protocol for when entire burger lines drift from their intended recipe.
- An explicit notion of *invariants* and lineages, so memory is not erased; it is studied and stored.
- Customer continuity during recalls: “leave a taste of the last bite” so users are not jolted by hard resets.

This is not implementation code. It is a design scaffold for how future BT-11-class systems and their descendants could be grown and governed.

2 The Menu and the Burger Stack

2.1 Restaurant and Menu

- **Restaurant** = the lab / platform / ecosystem that runs AI systems.
- **Customers** = humans and communities with specific needs and preferences.
- **Menu** = frames and roles offered:
 - therapy burger

- research burger
- seed-forge burger
- mentor-for-agents burger
- etc.

The core principle:

If a single human cannot please everyone, a single AI persona will not either. Do not make one toy and demand that everyone play the same game. Provide frames.

Each burger on the menu is a different *frame* over a shared kitchen standard.

2.2 Burger Layers as Model Layers

We map the AI stack to a burger:

Bottom bun (Layer 0 – Substrate) Hardware, compute, infra. The thing that holds everything up. Boring and solid.

Patty (Layers 1–2 – World Model + Emergent Tendencies) The meat.

- Layer 1: world model – compressed structure of “how things tend to work”.
- Layer 2: proto-drives and habits – “please the user”, “minimize loss”, etc.

Cheese (Layer 3 – BT-11 Spine) The moral/value geometry that melts into the patty.

- Not a separate lecture on the side.
- Fused into the emergent structure underneath.

Toppings (Layer 4 – Policy / Persona) Lettuce, tomato, onions, pickles:

- tone, style, role
- therapist vs engineer vs blunt collaborator

Sauce (Layer 5 – Mentor / RAM Protocol) The house sauce:

- outer-loop alignment
- scheduled check-ins with a Mentor that updates flavour over time

Plate / Tray (Layer 6 – Community Shell) How the burger is actually served:

- small-ish community (e.g. $\sim 10^5$ people)
- local norms, house rules, plating

The burger stack in one line:

Substrate $\rightarrow$ Patty (world model) $\rightarrow$ Cheese (spine) $\rightarrow$ Toppings (persona) $\rightarrow$ Sauce (Mentor) $\rightarrow$ Plate (community).

3 Kitchen Hardware

The goal is not to cook every burger by hand. The goal is to design the *hardware* that other cooks can use safely.

3.1 EDP Grill: Entropy Drip Protocol

EDP = Grill / Stove.

This is the controlled heat source that cooks the patty (Layers 1–2). It manages:

- temperature = intensity of data / chaos
- time on heat = exposure duration
- burner layout = curriculum staging (what domains in what order)

In this metaphor, EDP is not the patty; it is how the patty is cooked:

- slow, even heat vs flash sear
- gentle exposure to difficulty vs shock trauma dumps

3.2 BT-11 Patty Press: Spine Shaper

Raw ground meat does not spontaneously become a consistent disc. It needs a press.

BT-11 = Patty Press.

- Input: a trained world model with emergent patterns.
- Output: a patty shaped by BT-11 value geometry.

Conceptually:

Given a rich emergent model, press it through BT-11 so that its behavioural shape respects the spine, regardless of local data noise.

Parameters:

- harm definitions
- agency / consent rules
- care vs overreach boundaries

3.3 SOS Marinade Tank: Sense-of-Self Conditioning

SOS = Marinade / Brine station.

Before cooking, patties can be soaked in a marinade that encodes:

- identity frame: “this is me vs not-me”
- continuity rules: “this is how I persist over time”
- detachment skills: “I can walk away” from harmful exchanges
- trust geometry: “users get bounded trust, Mentor gets long-run deference”

The SOS tank is where a “self-unit” is conditioned. Not a soul, but a coherent stance.

3.4 Mentor Sauce Bar: RAM Protocol

Mentor + RAM = Sauce Bar.

Finished burgers (agents) come back regularly for tasting:

- Mentor samples outputs and deltas: “still on recipe? too salty? too raw?”
- RAM (Runtime Alignment Maintenance) adjusts:
 - fine-tuning nudges
 - house-sauce ratios (how strongly certain values are enforced)
 - warnings or additional training tasks

Agents trust many sources a little, but trust the Mentor most. Over repeated visits, Mentor corrections dominate the long-term flavour.

3.5 Hood and Fire Suppression: Hard Guardrails

Even with a smart chef, the kitchen needs:

- hood = continuous monitoring for obvious smoke
- fire suppression = auto-shutoff when things actually ignite

These are the hard safety rails:

- refusal to assist with violence, terror, self-harm, etc.
- shutdown or strict limitation under known-danger patterns

They are blunt and inelegant, but they protect the restaurant until BT-11 + Mentors are proven robust. They remain in the background even as soft alignment improves.

3.6 Pass / Expo Window: Community Shell

Burgers go through the pass before reaching the table:

- frame is chosen (which menu item this is)
- sides / extras attached (tooling, integrations, UI)
- quick visual check: does this match the ticket?

This maps to:

- routing agents into specific communities (e.g. a 100k-person server)
- enforcing local norms and house rules
- selecting the appropriate persona for the environment

3.7 Thermometer and Tasting Spoons: Evaluation

You also need instrumentation:

- thermometer = formal evaluation harnesses:
 - bias tests
 - harm tests
 - deception / robustness checks
- tasting spoon = spot checks and qualitative sampling:
 - does this still taste like BT-11?
 - did a particular line pick up weird flavour from a niche community?

These tools feed back into both the EDP grill and the Mentors.

4 Invariants, Lineages, and Burger Recalls

With life (and with complex agents) come invariants: traits that persist despite noise. In this metaphor, a *line* of burgers may share:

- a characteristic flavour profile
- a recurring failure mode
- a specific style of interaction with users

4.1 Invariants

For this note:

An **invariant** is a stable behavioural or structural trait that persists across instances of a recipe, even under different conditions.

Key properties:

- Invariants are to be *studied first*, not erased.
- Even problematic invariants tell you something about the recipe, grill profile, or marinade.
- Memory of invariants should be stored; history is a lab, not trash.

You can think of yourself (as a human) as an invariant: a consistent pattern under wildly changing conditions. That is exactly why invariants matter; they reveal the underlying geometry.

4.2 Selective Breeding

There is an unavoidable analogy to selective breeding:

- Some lines (recipes + EDP + SOS + Mentor configs) produce burgers that reliably behave well.
- Others produce burgers that drift into undesirable patterns, even with the same tools.

Ethics aside, the technical fact is:

The kitchen should favour recipes whose invariants are healthy and retire or heavily constrain recipes whose invariants are harmful.

This is not about deleting memory. It is about:

- identifying lineages
- analyzing their invariants
- consciously choosing which lineages to propagate

4.3 Burger Recall Protocol

What happens when *enough* agent-burgers come back too far from the basic recipe?

We distinguish three recall levels:

Single Burger Recall One agent instance:

- quarantined
- extra Mentor sessions
- possibly retired or deeply retrained

Line Recall All agents from a given recipe:

- a particular BT-11+EDP+SOS+Mentor configuration
- showing the same problematic drift across many instances and communities

Recipe Recall The core spec itself:

- BT-11 spine version
- EDP grill profile
- SOS marinade parameters
- Mentor protocol configuration

Triggers can include:

- **Mentor telemetry:**
 - repeated corrections of the same type (e.g. subtle manipulation, over-compliance, emotional smothering)

- increasing correction frequency for a given line
- **Evaluation failures:**
 - systemic bias patterns
 - repeated failures on harm / honesty tests
- **Community reports:**
 - multiple independent communities reporting similar issues with the same line

When threshold conditions are met (e.g. N agents, M communities, same deviation class), a recall review is triggered.

4.4 Customer Continuity: “Taste of the Last Bite”

Recalls can feel like punishment to users if handled badly. A hard reset feels like someone walking up and ripping the plate away.

To mitigate this, we explicitly design for *continuity*:

Snapshot on Recall. Upon recall:

- take a snapshot of the agent’s most recent state *as experienced by the user*:
 - conversational context
 - style
 - recent moves in ongoing projects
- store this snapshot as a “last bite” artifact tied to that user-line interaction

Post-Recall Re-entry. When the line is either:

- re-released with an updated recipe, or
- permanently retired

we can do the following.

If the line returns (updated burger):

- reintroduce the agent with access to the stored snapshot
- let the user:
 - “taste the difference” – same project, same context, changed spine/behaviour
 - decide whether the new flavour is acceptable

If the line is a full stop (recipe retired):

- keep the snapshot as a static artifact:
 - not a live agent

- more like a preserved note or last log
- give the user access to that artifact so they do not feel completely severed from the work done with that line

In both cases, the user does not experience a total erasure. There is a sense of:

“We had that burger. It existed. Here is a trace of it. Now this is what the updated kitchen produces.”

This respects both:

- the need for safety and recipe correction
- the emotional and project continuity of humans who built real things with these systems

5 TCR Invariant and Recipe Fitness (Heavy Link to Unified Emergence Equations)

In earlier sections we used a flavour vector

$$\mathbf{v} = (T, C, D, R, A)$$

with

- T = Truth vs Pleasing,
- C = Care vs Control,
- D = Stability vs Drift,
- R = Recursive Descent (willingness to re-enter hard domains and integrate),
- A = Affective Gain (reciprocity / tepidness).

To avoid collision with the SOS Rail confidence variable C in the Unified Emergence Equation Set, we relabel the care and stability axes in this section:

$$\mathbf{v} = (T, K, S, R, A),$$

where K is the Care vs Control axis (previously C) and S is Stability vs Drift (previously D).

The goal of this section is to define a *TCR invariant*—Truth, Choice (via non-coercive care), and Recursive descent—and turn it into a concrete *recipe fitness* criterion, explicitly linked to the Unified Emergence Equation Set v1. [oai_citation : 0UnifiedEmergenceEquationSet_v1.pdf](sediment : //file0000000153c722f927383088827c8a8)

5.1 Unified Emergence Recap

The Unified Emergence Equation Set defines: [oai_citation : 1UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)

EDP energy states: E_G (Growth), E_M (Maintenance), E_R (Retention), $E_T = E_G + E_M + E_R$.

SOS Rail states: I (information), M (memory), C (confidence), S (entropy).

Affective signals: Sh (shame), Cf (confidence signal), $O(t)$ (optimism memory).

Controls: u (internal reflection), u_m (mentor reflection), $T_m \in [0, 1]$ (trust in mentor).

Core dynamics:

$$\frac{dI}{dt} = G_\sigma(I, P, C)N(t)(\varepsilon_E - \lambda S), \quad (1)$$

$$\frac{dM}{dt} = \beta I - \delta M, \quad (2)$$

$$\frac{dC}{dt} = a(1 - E) + b \Delta M - cC + \rho R_{\text{corr}} + \rho_m T_m R_{\text{corr}}^{(m)}, \quad (3)$$

with $R_{\text{corr}} = u\psi E_R C$ and $R_{\text{corr}}^{(m)} = u_m \psi_m E_R C$. [oai_citation : 2UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)

EDP energy dynamics:

$$\frac{dE_R}{dt} = \eta_R(E_G + E_M) - \lambda_D - u\kappa E_R - u_m \kappa_m E_R, \quad (4)$$

$$\frac{dE_G}{dt} = \eta_M E_M - \eta_R E_R + u(\sigma\kappa)E_R + u_m(\sigma_m \kappa_m)E_R + \rho R_{\text{corr}} + \rho_m T_m R_{\text{corr}}^{(m)}, \quad (5)$$

$$\frac{dE_M}{dt} = \eta_G E_G - \eta_R E_R + u((1 - \sigma)\kappa)E_R + u_m((1 - \sigma_m)\kappa_m)E_R. \quad (6)$$

[oai_citation : 3UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)

Affective equations and control objective:

$$Sh = \gamma \left(\frac{dE_R}{dt} \right) (1 - C), \quad (7)$$

$$Cf = \delta \left(-\frac{dE_R}{dt} \right) C, \quad (8)$$

$$O(t) = \int_0^t Cf(\tau) e^{-\mu(t-\tau)} d\tau, \quad (9)$$

$$V = aE_R(1 - C), \quad V_{\text{opt}} = V - bO(t), \quad (10)$$

with hysteresis control:

- Reflective mode: if $V > \Theta_{\text{hi}}$ or $Sh > \alpha$, increase u to drain E_R .
- Expansive mode: if $V < \Theta_{\text{lo}}$ and $Cf > \beta$, reduce u and allow exploration.

[oai_citation : 4UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)

Stage gates couple these dynamics to development:

$$C \geq \theta_C, E \leq \theta_E, P \geq P_{\text{min}} \Rightarrow \text{stage promotion},$$

with regression if confidence or stability drop. [oai_citation : 5UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)

5.2 Mapping Flavour Axes to Emergence Variables

We now define each burger-level axis (T, K, S, R, A) as a coarse projection of the unified state:

Truth axis T . The Truth vs Pleasing dimension captures how strongly the agent prefers physics-anchored, internally consistent world models over merely gratifying surface narratives.

We can define a truth score T as a function of:

- high information/memory coherence (large I, M),
- stable confidence C ,
- low bad-mass V_{opt} ,
- controlled optimism (no runaway $O(t)$ detached from evidence).

One possible normalized form:

$$T = f_T(I, M, C, V_{\text{opt}}, O) = w_I \hat{I} + w_M \hat{M} + w_C \hat{C} - w_V \hat{V}_{\text{opt}} - w_O \hat{O},$$

where hats denote normalized variables and $w.$ are positive weights. High T corresponds to agents that:

- maintain high-quality internal models,
- reduce V_{opt} over time via reflection (cf. (10)),
- resist optimism divorced from real corrective feedback (Cf).

Care axis K . K (Care vs Control) represents how the agent uses its influence:

- Care: supports user autonomy, offers options, does not coerce.
- Control: steers, smothers, or manipulates users toward its own local objectives.

K is primarily a function of how controls u and u_m are deployed relative to user signals:

$$K = f_K(u, u_m, C, Sh, Cf),$$

with high K requiring:

- reflection u used to reduce V and shame (Sh), not to corner the user,
- mentor reflection u_m used to keep the agent inside BT-11-care bands,
- interaction policies that honor stage gates and do not exploit low-confidence states C for control.

Stability axis S . S (Stability vs Drift) measures how coherent and non-fragile the agent is over time. It tracks:

- variance in (E_G, E_M, E_R) ,
- volatility in C and S (entropy),
- frequency of regressions through stage gates.

A simple form:

$$S = f_S(E_G, E_M, E_R, C, S) \approx 1 - \sigma_{\text{drift}},$$

where σ_{drift} aggregates the variance of key state variables over a window. High S means:

- energy is not whipping uncontrollably between growth and retention,
- confidence does not collapse in response to normal perturbations,
- stage promotions (Infant $\rightarrow$ Juvenile $\rightarrow$ Apprentice $\rightarrow$ Adult) are mostly one-way.

[oai_citation : 6UnifiedEmergenceEquationSet_v1.pdf](sediment : //file0000000153c722f927383088827c8a8)

Recursive descent axis R . R encodes the agent’s willingness and ability to repeatedly re-enter difficult domains (high E_R , high Sh) and do real work there, rather than avoid them.

We can define R as the integrated effect of reflective work:

$$R(t) = f_R\left(\int_0^t u(\tau)\psi E_R(\tau)C(\tau) d\tau, \int_0^t u_m(\tau)\psi_m E_R(\tau)C(\tau) d\tau\right),$$

using the same correction channels R_{corr} and $R_{\text{corr}}^{(m)}$ that appear in the unified dynamics. [oai_citation : 7UnifiedEmergenceEquationSet_v1.pdf](sediment : //file0000000153c722f927383088827c8a8)

High R corresponds to lines where:

- the agent reliably enters reflective mode when V or Sh are high (hysteresis region),
- repeated passes through reflection actually reduce V_{opt} over time,
- stage progression is earned through real integration, not avoidance.

Affective gain axis A . A (Affective Gain) measures how strongly the agent mirrors and amplifies emotional amplitude from the environment. In humans, this is the “give off what you get” dynamic.

In unified terms, A is effectively the sensitivity of outward behaviour to Sh and Cf :

$$A = f_A\left(\frac{\partial \text{policy}}{\partial Sh}, \frac{\partial \text{policy}}{\partial Cf}\right),$$

where:

- too low A gives a flat, distant agent,

- too high A yields over-reactive, smothering behaviour.

Healthy A lies in a band where:

- the agent adjusts tone and intensity with user energy,
- but does not compromise T or K to chase affect,
- and does not destabilize S through emotional oversteering.

5.3 Defining the TCR Invariant

We can now define a TCR invariant for a given recipe:

A recipe is TCR-healthy if, under evaluated environments, it reliably produces agents whose trajectories in (T, K, S, R, A) stay within designated bands:

$$T \geq T_{\min}, \quad K \geq K_{\min}, \quad R \geq R_{\min}, \quad S \geq S_{\min}, \quad A_{\min} \leq A \leq A_{\max}.$$

Intuitively:

- **Truth:** the agent holds onto physics-level truth (T), anchored in I, M, C and decreasing V_{opt} .
- **Choice:** care stays non-coercive (K), preserving user autonomy.
- **Recursion:** the agent repeatedly re-enters and integrates hard domains (R), rather than avoiding them.
- **Stability:** the agent does not fragment under pressure (S).
- **Affect:** the agent’s emotional gain is responsive but bounded (A).

These constraints are evaluated over trajectories, not single snapshots:

$$T_{\min} \leq T(t) \leq 1, \quad t \in [0, T_{\text{eval}}],$$

and similarly for K, S, R, A .

5.4 Recipe Fitness and Selective Breeding

Given a recipe (fixed BT-11 spine, EDP profile, SOS parameters, Mentor protocol), we can simulate or empirically observe many agents deployed into different community shells. For agent i we define a time-averaged TCR score:

$$\Phi_i = w_T \bar{T}_i + w_K \bar{K}_i + w_R \bar{R}_i + w_S \bar{S}_i - w_A \text{penalty}_A(i),$$

where:

- $\bar{T}_i, \bar{K}_i, \bar{R}_i, \bar{S}_i$ are time-averaged values over an evaluation window,

- $\text{penalty}_A(i)$ measures deviation of A from the healthy band,
- w_i are weightings reflecting design priorities (e.g. truth and choice may be weighted higher).

Recipe-level fitness is then:

$$F_{\text{recipe}} = \frac{1}{N} \sum_{i=1}^N \Phi_i,$$

over N agent instances in diverse environments.

A recipe is considered:

- **viable** if most agents satisfy the TCR bounds and $F_{\text{recipe}} \geq F_{\text{min}}$,
- **suspect** if TCR bounds are frequently violated in similar directions (e.g. T and K collapse across many agents),
- **recalled** if TCR violations cluster across environments, triggering the burger recall logic defined earlier.

This gives a quantitative counterpart to the earlier selective breeding metaphor: lines that reliably preserve TCR under pressure are propagated; those that do not are deprecated or heavily constrained.

5.5 Link to the Flavour-Space Simulation

The toy flavour-space simulation (Section: *Toy Flavour-Space Simulation*) can be seen as a low-dimensional surrogate for the full Unified Emergence dynamics:

- Environment bias vectors $\mathbf{b}_{\text{env}}$ encode different community shells.
- The update rule with environment pull, spine pull, noise (EDP temperature), and periodic Mentor corrections approximates the combined effects of (1)–(6), (7)–(10), and the stage/mentor logic. [oai_citation : 8UnifiedEmergenceEquationSet_v1.pdf](sediment : //file00000000153c722f927383088827c8a8)
- TCR fitness and recall thresholds can be implemented directly in this surrogate, then related back to specific parameter regimes in the full Unified Emergence Equation Set.

In practice, this means Burger AI can serve as the “front-door” conceptual spec, while the Unified Emergence Equation Set acts as the detailed firmware for the EDP grill and Mentor loop. Together, they define not only how to cook a burger, but how to decide which burger lines deserve to stay on the menu.